

Homework 8

Problem 1. Let π be a random permutation of the set $\{1, 2, \dots, n\}$.

- a) Determine the expected length of the cycle of π containing the number 1.
- b) Determine the expected number of cycle of π .

Solution.

- a) Let A_i be the event that 1 lies in a cycle of length i , then $P(A_i) = (n-1)^{i-1} \cdot \frac{(n-i)!}{n!} = \frac{1}{n}$. Therefore the expectation will be $\sum_{i=1}^n i \times \frac{1}{n} = \frac{n+1}{2}$.
- b) Let E_n be the expected number of cycle over a set with n vertices. Then consider the length of the cycle containing 1, according to 1) we have that $E_n = \sum_{k=1}^n (E_{n-k} + 1) \times \frac{1}{n}$. i.e., $n \cdot E_n = \sum_{i=0}^{n-1} E_i + n$. Therefore $E_n = E_{n-1} + \frac{1}{n}$. Finally, $E_n = \ln n$.

An alternative solution: Let A_{ij} be the event ' j lies in a cycle of length i '. By a), we have $P(A_{ij}) = \frac{1}{n}$. Let the random variable $I_{A_{ij}}$ be 1 when A_{ij} happens. The resulting expectation then equals to $\sum_{i,j=1}^n \frac{1}{i} \mathbf{E}(I_{A_{ij}}) = \sum_{i,j=1}^n \frac{1}{i} \cdot \frac{1}{n} = \frac{1}{n} \sum_{j=1}^n \sum_{i=1}^n \frac{1}{i} = \frac{1}{n} \cdot n \cdot \ln n = \ln n$.

□

Problem 2. A permutation on the numbers $\{1, 2, \dots, n\}$ can be represented as a function $\pi : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$. Now the permutation π is chosen uniformly at random from all permutations. Recall that a fixed point of a permutation π is a value for which $\pi(x) = x$. We use $Fix(\pi) = \{x \mid \pi(x) = x\}$ to represent the set of fixed point of π . Find the expectation and variance of the size of $Fix(\pi)$, i.e., $\mathbf{E}[|Fix(\pi)|]$, $\mathbf{Var}[|Fix(\pi)|]$.

Solution. For given permutation π , to every $i \in \{1, 2, \dots, n\}$, define random variable x_i as $x_i = 1$ if $i \in Fix(\pi)$, and $x_i = 0$ if $i \notin Fix(\pi)$. Then $|Fix(\pi)| = \sum_{i=1}^n x_i$.

To the expectation: $\mathbf{E}[|Fix(\pi)|] = \sum_{i=1}^n 1 \cdot \Pr(f(i) = i) = \sum_{i=1}^n 1 \cdot \frac{1}{n} = 1$.

To the variance:

$$\begin{aligned}
\mathbf{Var}[|Fix(\pi)|] &= \mathbf{E}[|Fix(\pi)|^2] - \mathbf{E}[|Fix(\pi)|]^2 \\
&= \mathbf{E}[(\sum_{i=1}^n x_i)^2] - 1 \\
&= \mathbf{E}[\sum_{i=1}^n x_i^2] + \mathbf{E}[\sum_{1 \leq i \neq j \leq n} x_i x_j] - 1 \\
&= \sum_{i=1}^n \mathbf{E}[x_i^2] + \sum_{1 \leq i \neq j \leq n} \mathbf{E}[x_i x_j] - 1 \\
&= \sum_{i=1}^n 1 \cdot \Pr(x_i = 1) + \sum_{1 \leq i \neq j \leq n} 1 \cdot \Pr(x_i = x_j = 1) - 1 \\
&= n \cdot \frac{1}{n} + n(n-1) \frac{(n-2)!}{n!} - 1 \\
&= 1
\end{aligned}$$

□

- Problem 3.
1. Prove that, for every integer n , there exists a coloring of the edges of the complete graph K_n by two colors so that the total number of monochromatic copies of K_4 is at most $\binom{n}{4} 2^{-5}$.
 2. Give a randomized algorithm for finding a coloring with at most $\binom{n}{4} 2^{-5}$ monochromatic (i.e. single-color) copies of K_4 that runs in expected time polynomial in n .
 3. Show how to construct such a coloring deterministically in polynomial time using the method of conditional expectations. (*)

Solution.

1. Coloring every edge in K_4 by red or blue with probability $1/2$. The expected value of the total number of monochromatic copies of K_4 is then $2 \times \binom{n}{4} \times \left(\frac{1}{2}\right)^6$. Then there must exist some coloring scheme where the total number of monochromatic copies of K_4 is less or equal to $\binom{n}{4} 2^{-5}$ (otherwise the expectation would be strictly larger than $\binom{n}{4} 2^{-5}$).
2. Color each edge independently and uniformly. Let $p = \Pr(X \leq \binom{n}{4} 2^{-5})$ where X is the number of chromatic K_4 .

$$\begin{aligned}
\binom{n}{4} 2^{-5} &= \mathbf{E}(X) \\
&= \sum_{i \leq \binom{n}{4} 2^{-5}} i \cdot \Pr(X = i) + \sum_{i > \binom{n}{4} 2^{-5}} i \cdot \Pr(X = i) \\
&\geq p + (1-p) \left(\binom{n}{4} 2^{-5} + 1 \right)
\end{aligned}$$

which implies $p \geq \frac{32}{\binom{n}{4}}$. The expected number of sampling before finding a suitable coloring is $1/p = \frac{\binom{n}{4}}{32}$. For each sampling, the time needs

to count the number of chromatic K_4 is bounded by $\binom{n}{4}$ which is also polynomial. Thus the expected running time of this algorithm is polynomial.

3. Assign colors to each edge of K_n deterministically, in any order $x_1, x_2, \dots, x_m$ (where x_i is the random variable of the coloring of edge i). Suppose that we have assigned the first k edges. Let $y_1, y_2, \dots, y_k$ be the corresponding assigned values. We compute the quantities:

$$\mathbf{E}[X \mid x_1 = y_1, x_2 = y_2, \dots, x_k = y_k, x_{k+1} = \text{red}];$$

$$\mathbf{E}[X \mid x_1 = y_1, x_2 = y_2, \dots, x_k = y_k, x_{k+1} = \text{blue}];$$

and then choose the setting with the smaller expectation.

□